인공지능수학기초 (01)

Final Exam
Dec. 17, 2024

1. (4+4+4 pts) Continuous random variable X가 다음의 분포를 갖는다.

$$f(x) = \alpha \exp(-|x|/\sigma),\ -\infty < x < \infty,\ \sigma > 0$$

   (a) α를 구하시오.
   (b) $E(X)$를 구하시오.
   (c) $Var(X)$를 구하시오.

2. (4+4 pts) 다음의 geometric 분포를 보자.

$$f_X(x) = pq^{x-1} \quad x \ge 1,\ 0 < p, q < 1,$$

   (a) MGF $E(e^{tX})$를 구하시오.
   (b) (a)를 이용해서, $E(X)$를 구하시오.

3. (4+6 pts) Bivariate random variable X와 Y의 joint pmf가 다음과 같다.

$$f_{X,Y}(x,y) = \binom{n}{x}\binom{m}{y} p^{x+y}(1-p)^{m+n-x-y},\ 0 \le x \le n,\ 0 \le y \le m,\ 0 < p < 1.$$

   (a) X와 Y의 marginal pdf를 각각 구하시오.
   (b) $X+Y$의 분포를 구하시오.

4. (4+6 pts) 함수 $g(x)$ 가 $-\infty < E(g(x)) < \infty$ & $-\infty < g(-1) < \infty$ 라고 하자. Random variable X 가 다음과 같이 Poisson 분포를 따를 때, 다음을 보자.

$$f_X(x) = \frac{\lambda^x}{x!} e^{-\lambda},\ (x = 0, 1, 2, \cdots).$$

   (a) 다음이 성립함을 보이시오.

$$E(\lambda g(X)) = E(Xg(X-1)).$$

   (b) (a)를 이용해서, $E(X^2)$을 구하시오.

1. $f(-x) = f(x)$,

a) $\therefore \int_0^\infty \alpha \exp\left(-\frac{x}{\sigma}\right) dx = 1/2$.

$= -\sigma\alpha \exp\left(-\frac{x}{\sigma}\right)\Big|_0^\infty = \sigma\alpha = 1/2 \quad \therefore \alpha = \frac{1}{2\sigma}$.

b) $E(X) = \int_{-\infty}^{\infty} \frac{1}{2\sigma} x \exp\left(-\frac{|x|}{\sigma}\right) dx \leftarrow$ odd func.

$= 0$

c) $E(X^2) = \int_{-\infty}^{\infty} \frac{1}{2\sigma} x^2 \exp\left(-\frac{|x|}{\sigma}\right) dx$

$= \frac{1}{\sigma}\int_0^\infty x^2 \exp\left(-\frac{x}{\sigma}\right) dx \Leftarrow \frac{x}{\sigma} = y,\ x = \sigma y,\ dx = \sigma dy$

$= \frac{1}{\sigma}\int_0^\infty \sigma^2 y^2 e^{-y} \sigma dy$

$= \sigma^2 \Gamma(3) = 2\sigma^2$.

$\therefore Var(X) = E(X^2) - (EX)^2$

$= 2\sigma^2 - 0 = 2\sigma^2$.

2. a) $E(e^{tx}) = \sum_{x=1}^{\infty} e^{tx} p q^{x-1}$

$= \sum_{x=1}^{\infty} \frac{p}{q}(e^t q)^x = \frac{p}{q} \cdot \frac{e^t q}{1-e^t q} = \frac{e^t p}{1-e^t q}$

$(|e^t q| < 1 \rightarrow t < -\ln q)$

없으면 1점 감점.

b) $E(X) = \frac{d}{dt} E(e^{tx})\Big|_{t=0}$

$$= \frac{(1-e^t q)e^t p + e^t q e^t p}{(1-e^t q)^2} \Big|_{t=0}$$

$$= \frac{p}{(1-q)^2} = \frac{1}{p}$$

3. a) factorizable $\rightarrow$ independent.

$$\therefore f_X(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$f_Y(y) = \binom{m}{y} p^y (1-p)^{m-y}$$

b) let $Z = X+Y$, then

$$f_Z(z) = f_X(z) * f_Y(z)$$

$$= \sum_{k=-\infty}^{\infty} f_X(k) f_Y(z-k) \quad \left(\begin{matrix} k \geq 0 \\ z-k \geq 0 \end{matrix} \right.$$

$$= \sum_{k=0}^{z} \binom{n}{k} p^k (1-p)^{n-k} \binom{m}{z-k} p^{z-k} (1-p)^{m-z+k}$$

$$= \sum_{k=0}^{t} \binom{n}{k} \binom{m}{z-k} p^z (1-p)^{m+n-z}$$

$$= \binom{m+n}{z} p^z (1-p)^{m+n-z}$$

$$\longrightarrow \mathrm{bin}(m+n, p)$$

4. a) $E(Xg(X-1)) = \sum_{x=0}^{\infty} x g(x-1) \frac{\lambda^x}{x!} e^{-\lambda}$

$$= \sum_{x=1}^{\infty} g(x-1) \frac{\lambda^x}{(x-1)!} e^{-\lambda}, \quad \left(\begin{array}{l} x-1=y \\ y=x+1 \end{array} \right.$$

$$= \sum_{y=0}^{\infty} g(y) \frac{\lambda^{y+1}}{y!} e^{-\lambda}$$

$$= \sum_{y=0}^{\infty} \lambda g(y) \cdot \frac{\lambda^y}{y!} e^{-\lambda} = E(\lambda g(x))$$

b) let $g(x) = x$, then.

$$E(\lambda X) = E(X(X-1))$$

$$\lambda E(x) = E(X^2) - E(X)$$

$$\therefore E(X^2) = (\lambda+1) E(X).$$

let $g(x) = 1$, then

$$E(\lambda) = E(X) \quad \therefore E(x) = \lambda.$$

$$\therefore E(X^2) = (\lambda+1)\lambda = \lambda^2 + \lambda.$$